NanoAgent Plugins: Fundamentals and How to Build One

NanoAgent 0.5.0

1 What a plugin is

An agent saying it has finished a task does not show that its result is right. Judging an agent takes a precise task, an environment anyone can reproduce, and a verifier that checks the actual result. A NanoAgent plugin packages all three: typed tools served over MCP, tasks built from them, a deterministic verifier, and a front end, NanoAgent's chat, which runs the tasks and shows each result with its verdict. NanoAgent loads a plugin only when every `nanoagent.plugin/v1` rule below holds, rechecks them on every tool call, and lists every broken rule at once.

2 The fundamentals

1. **Typed in, typed out.** Every tool and task has an input class and an output class. They are the only shapes that cross the plugin's boundary, and every value is built into its class on the way in and out.
2. **Decisions are typed questions.** The model chooses: the next tool, and for each field an allowed value or an existing value of the same type. It does not invent values. Any model that answers typed questions can be used, and the contract does not depend on which.
3. **Rules in code, judgment in the model.** If a value follows from rules, a tool computes it. The model decides only what needs judgment.
4. **Every result is verified by code. LLM as judge is banned.** The verifier is plain, sandboxed, deterministic Python that decides every case. A task whose correctness code cannot check does not belong in a plugin.
5. **Natural language only at the end,** to explain the typed result to the person.

How questions are batched, what background the model sees and whether it is cached are harness settings. They change without touching any contract.

3 Anatomy

A plugin folder holds exactly these files, and nothing else at its top level. The Hello World starter has the smallest working example of each, and the arXiv Source plugin a full one, with network access, five tasks and a verifier module per tool.

File	What it holds
`contract.py`	Declarations only: `Tool` and `Task` Enums, slimconfig `Output` classes, and the `TOOLS` and `TASKS` tables pairing each with its input and output classes. Tools are snake_case, tasks kebab-case, and class names end in `Input` or `Output`. Field descriptions become the JSON Schema everyone sees.
`nanoagent.yaml`	The manifest: id; `version` (PEP 440, the same as `pyproject.toml`'s); `status`, `ready` or `in-development` (listed, never run); the one NanoAgent minor version it runs on, `~=X.Y.0`; the runtime, always `uv run --frozen python -m plugin`; permissions (hosts it may reach, where it may write); and the tools and tasks tables. Each task has a `runner`, `tools`, `reference` and `instructions`.
`src/`	The tools and the MCP server. A tool takes its input class, returns its output class, and raises on failure. It writes only to its workspace.

File	What it holds
`verifier/`	`verify_input` and `verify_output` run on every tool call, and `verify_task` runs once on the task's output. Each returns `valid`, `score` (0 to 1), `message` and `details`, and checks what a value means, never its shape. Pure standard-library imports only, no file writes, sandboxed without network, and run twice: the verdicts must match.
`tests/`	The plugin's tests, including `test_verifier.py`.
`ui/`	`renderers.js` and `renderers.css`. Outputs are drawn from their JSON Schema unless `renderers.js`'s default export, keyed by a task's output class, draws them. Renderers import only relative paths (the kit and the plugin's own `ui/` files), never touch the page, the window or the network, and their CSS applies only inside the output.
`product/`	`product.yaml` (tagline, summary, 2 to 4 highlights, 2 to 5 steps, a `ready` plugin's all its own) and `icon.svg` (at most 64 KB, drawing only), the catalog page.
`run.sh`, `.env`	`run.sh` exits 0 when this machine can run the plugin. `.env` lists variable names, never values.
Standard files	`README.md`, `LICENSE`, `pyproject.toml`, `uv.lock` and `.gitignore`, the template's. The README has the sections Tools, Layout, and Develop and run, and names every tool and task. `pyproject.toml` names the project `nanoagent-<id>`, keeps the template's Python version and dependencies, and bounds every other dependency above and below.

4 How a task runs

A task's `runner` says who acts:

- `tool`: no model. One tool is called with the task's input, and its output is the task's, as in the starter's `say-hello`.
- `agent`: an LLM loop with the task's tools, ending with `submit`, which takes the task's output class, as in arXiv Source's `find-and-save-paper`.
- `decide`: the model only chooses: the next tool or `submit`, an Enum or `bool` member, or an existing value of exactly a field's type from the task's input or an earlier output of one of its tools, never the reference tool's. NanoAgent calls the tools and builds every value. Every answer is checked against its options, and a task with a field that cannot be filled this way is refused when the plugin loads.

A task's `reference`, when set, is a tool outside the task's tools. NanoAgent runs it first, out of the model's sight, and pins its output as the ground truth for `verify_task`, as arXiv Source's `write-daily-digest` pins `list_daily_papers` and `its verifier` checks the digest against it. Otherwise it is `null`.

5 Building your own plugin

Install Git, Python 3.12 or later and `uv`, then start from the starter, replacing `my-task` with your plugin's name:

```
git clone https://github.com/zeyuyang8/mochi-hello-world.git my-task && cd my-task
```

1. **Rename it** in `nanoagent.yaml`, the README's title, `LICENSE` and `pyproject.toml` (`nanoagent-<id>`) together, and keep `status: in-development` until it is done.
2. **Declare the types** in `contract.py`. Use Enums and `bool`s for anything that is a choice.
3. **Write the tools** in `src/plugin/`, one module per tool, and register each in `__main__.py`. Put every rule-based computation in a tool.
4. **Describe the tools and tasks** in `nanoagent.yaml`, and grant only the permissions the plugin needs.

5. **Write the verifier** and its tests: deterministic code that decides every case, never a model. Test correct, incorrect, missing, malformed and boundary results, not only the correct one.
6. **Finish the front end**: renderers only where the default drawing falls short, plus `product.yaml` and `icon.svg` in your own words, then set `status: ready`.
7. **Test it** with `uv sync --extra dev` and `uv run pytest`, and keep `uv.lock` current. Keep reports and run outputs outside the plugin folder.

The `starter's README` says how to add a tool or a task, and arXiv Source shows each piece at full size. The manifest's `nanoagent` field, such as `~=0.5.0`, is the one NanoAgent minor version the plugin runs on; NanoAgent raises it whenever a plugin written for the last one could break.

6 Publishing on RUSH Lab

1. Push the complete, tested plugin to a public GitHub repository.
2. Sign in at `https://rush-lab.org` and open the course workspace, `https://rush-lab.org/w/comp-480-580`. Add an OpenRouter API key and choose the model under **Settings** → **AI**.
3. In the workspace's **Marketplace**, select **+ Publish a plugin**, paste the repository URL (for a subfolder, `https://github.com/<owner>/<repo>/tree/<branch>/<folder>`), and select **Publish**. Fix any reported errors and publish again.
4. Once a course workspace owner adds it, open the plugin, ask the chat for the task, and check the verdict on its result card. **Passing the publication checks shows the format, not that the task or verifier is correct**: the verifier's tests and the runs' verdicts show that.

7 License and submission (COMP 480/580)

Every plugin carries the template's MIT `LICENSE`, with you as the copyright holder, and NanoAgent refuses a plugin without it. Publishing on RUSH Lab needs a public GitHub repository, so a published plugin is open source under that license.

Giving RUSH Lab permission to open-source your work is optional and does not affect grading. If you agree, include this statement with your submission:

> I voluntarily agree that my original contribution to [Project Name] may be open-sourced under the MIT License. This permission is optional and does not affect grading.

To keep your work confidential, do not publish it: email your submission to [instructor email] and copy [TA emails] instead.